

Spin part 1

Spin matrices are given by

$$S_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad S_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad S_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Given the eigenvectors

$$\begin{aligned} |x_+\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} & |y_+\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} & |z_+\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ |x_-\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} & |y_-\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} & |z_-\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{aligned}$$

spin matrices can also be written as

$$\begin{aligned} S_x &= \frac{\hbar}{2} |x_+\rangle \langle x_+| - \frac{\hbar}{2} |x_-\rangle \langle x_-| \\ S_y &= \frac{\hbar}{2} |y_+\rangle \langle y_+| - \frac{\hbar}{2} |y_-\rangle \langle y_-| \\ S_z &= \frac{\hbar}{2} |z_+\rangle \langle z_+| - \frac{\hbar}{2} |z_-\rangle \langle z_-| \end{aligned}$$

Eigenmath script